

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % ifconfig en0 | grep "inet "
    inet 10.7.28.24 netmask 0xffffff00 broadcast 10.7.31.255
adityasrivastava@Adityas-MacBook-Pro-103 ~ % networksetup -getinfo Wi-Fi

DHCP Configuration
IP address: 10.7.28.24
Subnet mask: 255.255.224.0
Router: 10.7.0.1
Client ID:
IPv4: Automatic
IPv4 IP address: none
IPv4 Router: none
Wi-Fi ID: 10:9f:41:c2:53:b9
adityasrivastava@Adityas-MacBook-Pro-103 ~ % netstat -nr | grep default
default      10.7.0.1          UGdg          en0
default      fe80::Nutun0      UGcIg         utun0
default      fe80::Nutun1      UGcIg         utun1
default      fe80::Nutun2      UGcIg         utun2
default      fe80::Nutun3      UGcIg         utun3
adityasrivastava@Adityas-MacBook-Pro-103 ~ % ping -c 4 10.7.9.8
PING 10.7.9.8 (10.7.9.8): 56 data bytes
64 bytes from 10.7.9.8: icmp_seq=0 ttl=64 time=20.645 ms
64 bytes from 10.7.9.8: icmp_seq=1 ttl=64 time=16.439 ms
64 bytes from 10.7.9.8: icmp_seq=2 ttl=64 time=42.553 ms
64 bytes from 10.7.9.8: icmp_seq=3 ttl=64 time=88.285 ms

--- 10.7.9.8 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 16.439/41.956/88.285/28.632 ms
adityasrivastava@Adityas-MacBook-Pro-103 ~ % ping -c 4 10.7.11.140
PING 10.7.11.140 (10.7.11.140): 56 data bytes
64 bytes from 10.7.11.140: icmp_seq=0 ttl=64 time=24.043 ms
64 bytes from 10.7.11.140: icmp_seq=1 ttl=64 time=47.733 ms
64 bytes from 10.7.11.140: icmp_seq=2 ttl=64 time=71.747 ms
64 bytes from 10.7.11.140: icmp_seq=3 ttl=64 time=109.628 ms

--- 10.7.11.140 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 24.043/63.263/109.628/31.690 ms
adityasrivastava@Adityas-MacBook-Pro-103 ~ % ping -c 4 10.7.30.18
PING 10.7.30.18 (10.7.30.18): 56 data bytes
64 bytes from 10.7.30.18: icmp_seq=0 ttl=64 time=105.944 ms
64 bytes from 10.7.30.18: icmp_seq=1 ttl=64 time=32.753 ms
64 bytes from 10.7.30.18: icmp_seq=2 ttl=64 time=66.122 ms

--- 10.7.30.18 ping statistics ---
4 packets transmitted, 3 packets received, 25.0% packet loss
round-trip min/avg/max/stddev = 32.753/68.200/105.964/29.927 ms
adityasrivastava@Adityas-MacBook-Pro-103 ~ % brew install nginx
  Auto-updating Homebrew...
Adjust how often this is run with '$HOMEBREW_AUTO_UPDATE_SECS' or disable with
'$HOMEBREW_NO_AUTO_UPDATE=1'. Hide these hints with '$HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').
=> Auto-updated Homebrew!
Updated 3 taps (mongodb/brew, homebrew/core and homebrew/cask).

  New Formulae
bend: language that blocks AI mistakes via proof
cmux: Run all your AI coding agents in tmux
confluence-markdown-exporter: Export Atlassian Confluence pages as Markdown files
go-srch-lint: Architecture linter for Go projects
hidapi-tester: Command-line tool to exercise USB HID devices via HIDAPI
kubectrl-radar: Missing open-source Kubernetes UI with a built-in MCP server for AI agents
libkivanc: VANC Processing Framework
mongodb/brew/mongodb-community@8.3
mongodb/brew/mongodb-enterprise@8.3
mongodb/brew/mongodb-mongocrypt@08.3
persona-server@08.4: Drop-in MySQL replacement
plink1: Whole-genome association analysis toolset
ratex: Fast TeX engine written in Rust
smclient-ng: Fast and user friendly way to interact with SMB shares
sofka: Kubernetes TUI, reimaged in Rust
somaliier: Relatedness, QC and ancestry checks from BAM/CRAM/VCF
taoup: Tao of Unix Programming with Ruby-powered ANSI-colored Fortunes
tcltls: OpenSSL extension to Tcl
tensai: Render Mermaid diagrams in the terminal
vortex: Terminal UI for WireGuard and OpenVPN with live telemetry and leak guarding
worktmux: Git worktrees + tmux windows for zero-friction parallel dev
xgcmmar: Structured generation and reasoning engine for LLMs
```

```
workreux: Git worktrees + tmux windows for zero-friction parallel dev
xgrammar: Structured generation and reasoning engine for LLMs
==> New Casks
activitywatch@experimental: Time tracker
agentloop-agent: Autonomous agent
amb-app: Coding agent and development environment
bettermacwidgets: Live-animated widgets that sit on the desktop
bitmuse: Bit-perfect music player for local hi-res libraries
buholaunchpad: Launchpad alternative
caldera-amp: Headless player for Plex music servers
calyx: Terminal for running and supervising coding agents
cline-app: AI coding agent
coland: Markdown editor
devicehall: Scanner for devices, open ports and security risks on the local network
distill-web-monitor: Monitor webpages for changes
doubsoine: Chinese input method with voice input and intelligent suggestions
font-google-sans
font-google-sans-flex
font-gennatander
font-tiny6-duo
font-xmind
ghostex: Workspace for running and reviewing multiple CLI coding agents
gitdifferent: Git client built around a three-way merge tool
gloomberb: Finance terminal
highball: Tool to run Windows games
hol-guard: Local-first runtime firewall for AI coding agents
hoptodesk: Remote desktop and remote support tool with end-to-end encryption
inoy: Proxy client
jlicone: Desktop client for JLCPCB quoting, ordering and order tracking
jpk1: Digital signature and authentication client for the My Number Card (JPK1)
kaku: Terminal optimised for AI coding
kktern: Local-first administration workspace for terminals, SSH, and SFTP
langdock: Platform for AI Adoption
local-studio: Local workstation for running and managing language models
look: Keyboard-first local launcher
markpad: Markdown viewer and editor
mdhero: Markdown viewer and editor
mindroom: Self-hostable AI stack for multi-user, multi-agent workflows on Matrix
muse: AI assistant for managing tasks, projects, and long-term goals
network-weather: Network diagnostics tool
nuphyio: Keyboard configurator for NuPhy devices
nvidia-pair: Local inference router for a group of compatible computers
openclip: Instant actions for selected text
opendisk: Disk space analyser
openplc-editor: IDE for creating programs for the OpenPLC Runtime
picmal: Converts and compresses images, video, audio and PDFs locally
plans: Project management tool for tasks, sprints and docs
recordly: Creator-focused screen recorder with auto-zoom, cursor effects, and more
relay: Menu bar app for building LLM prompts from files, clipboard and voice notes
reolink: Client for viewing and managing security cameras and NVRs
robbietilton-compositor: Photoshop alternative
ruswitcher: Keyboard layout switcher
saaga: Dictation app for Swedish and Norwegian
sindex: Code editor based on VS Code and Tauri
snapline: Screenshot, screen recording and GIF tool
ssl-368: Plugin mixer and control software for SSL 368-enabled hardware
ssl-download-manager: Software manager for SSL plug-ins
tapmap: Visualise network connections on an interactive world map
telarii: Markdown reader built for typography
telegram@bets: Messaging app with a focus on speed and security
tencent-yinyongheos: Tenant application store
tungsten-edge: Window-oriented taskbar that replaces the Dock
yap-app: On-device voice dictation

You have 27 outdated formulae and 1 outdated cask installed.

==> Downloading bottle manifests
✓ Bottle Manifest nginx (1.31.6) Downloaded 34.8KB/ 34.8KB
==> Would install 1 formulae:
nginx 1.31.6
Would install 1 dependency for nginx:
openssl@4
==> Would upgrade 2 dependencies for nginx:
ca-certificates
pcre2
==> Do you want to proceed with the installation? [y/n]
==> Fetching downloads for: nginx
✓ Bottle Manifest ca-certificates (2026-09-25) Downloaded 8.1KB/ 8.1KB
```

```
relay: Menu bar app for building LLM prompts from files, clipboard and voice notes
reolink: Client for viewing and managing security cameras and NVRs
robbleilton-compositor: Photoshop alternative
ruswitcher: Keyboard layout switcher
sagep: Dictation app for Swedish and Norwegian
sidx: Code editor based on VS Code and Tauri
snapline: Screenshot, screen recording and GIF tool
ssl-360: Plugin mixer and control software for SSL 360-enabled hardware
ssl-download-manager: Software manager for SSL plug-ins
tmap: Visualize network connections on an interactive world map
telari: Markdown reader built for typography
telegram@beta: Messaging app with a focus on speed and security
tencent-yingyongbao: Tencent application store
tungsten-edge: Window-oriented taskbar that replaces the Dock
yap-app: On-device voice dictation

You have 27 outdated formulae and 1 outdated cask installed.

=> Downloading bottle manifests
✓ Bottle Manifest nginx (1.31.6)
=> Would install 1 formula:
nginx 1.31.6
=> Would install 1 dependency for nginx:
openssl@4
=> Would upgrade 2 dependencies for nginx:
ca-certificates
pcr2
=> Do you want to proceed with the installation? [y/n]
=> Fetching downloads for: nginx
✓ Bottle Manifest ca-certificates (2026-09-25)
✓ Bottle Manifest pcr2 (10.49)
✓ Bottle Manifest openssl@4 (4.0.2.1)
✓ Bottle ca-certificates (2026-09-25)
✓ Bottle nginx (1.31.6)
✓ Bottle pcr2 (10.49)
✓ Bottle openssl@4 (4.0.2.1)
=> Installing nginx dependency: openssl@4
=> Upgrading openssl@4 dependency: ca-certificates
=> Pouring ca-certificates--2026-09-25.all.bottle.tar.gz
📦 /opt/homebrew/Cellar/ca-certificates/2026-09-25: 6 files, 204.4KB
=> Installing openssl@4
=> Pouring openssl@4--4.0.2.1.arm64.sequoia.bottle.tar.gz
Unlinking /opt/homebrew/Cellar/openssl@4/4.0.2.1: 6570 symlinks removed.
📦 /opt/homebrew/Cellar/openssl@4/4.0.2.1: 6,818 files, 23.8MB
=> Installing nginx
=> Pouring nginx--1.31.6.arm64.sequoia.bottle.1.tar.gz
📦 /opt/homebrew/Cellar/nginx/1.31.6: 28 files, 2.8MB
=> Caveats
=> nginx
Docroot is: /opt/homebrew/var/www

The default port has been set in /opt/homebrew/etc/nginx/nginx.conf to 8080 so that
nginx can run without sudo.

nginx will load all files in /opt/homebrew/etc/nginx/servers/.

To start nginx now and restart at login:
brew services start nginx
Or, if you don't want/need a background service you can just run:
/opt/homebrew/opt/nginx/bin/nginx -g daemon off!;
adityasrivastava@Adityas-MacBook-Pro-103 ~ % nginx -v
nginx version: nginx/1.31.6
adityasrivastava@Adityas-MacBook-Pro-103 ~ % nginx -t
nginx: the configuration file /opt/homebrew/etc/nginx/nginx.conf syntax is ok
nginx: configuration file /opt/homebrew/etc/nginx/nginx.conf test is successful
adityasrivastava@Adityas-MacBook-Pro-103 ~ % brew --prefix nginx
/opt/homebrew/opt/nginx
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo lsof -nP -iTCP:80 -sTCP:LISTEN
Password:
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo lsof -nP -iTCP:443 -sTCP:LISTEN
adityasrivastava@Adityas-MacBook-Pro-103 ~ % curl http://10.7.9.8:3001
{"message": "Backend A is running"}
adityasrivastava@Adityas-MacBook-Pro-103 ~ % curl http://10.7.11.140:3002
{"message": "Backend B is running"}
adityasrivastava@Adityas-MacBook-Pro-103 ~ %
```

```
Last login: Tue Sep 29 13:42:52 on ttys001
adityasrivastava@Adityas-MacBook-Pro-103 ~ % ipconfig getifaddr en0
10.7.28.24
adityasrivastava@Adityas-MacBook-Pro-103 ~ % ping -c 4 10.7.30.18
PING 10.7.30.18 (10.7.30.18): 56 data bytes
64 bytes from 10.7.30.18: icmp_seq=0 ttl=64 time=90.055 ms
64 bytes from 10.7.30.18: icmp_seq=1 ttl=64 time=8.807 ms
64 bytes from 10.7.30.18: icmp_seq=2 ttl=64 time=755.162 ms
64 bytes from 10.7.30.18: icmp_seq=3 ttl=64 time=6.046 ms

--- 10.7.30.18 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 6.046/215.018/755.162/313.673 ms
adityasrivastava@Adityas-MacBook-Pro-103 ~ % ping -c 4 10.7.9.8
PING 10.7.9.8 (10.7.9.8): 56 data bytes
64 bytes from 10.7.9.8: icmp_seq=0 ttl=64 time=23.481 ms
64 bytes from 10.7.9.8: icmp_seq=1 ttl=64 time=7.215 ms
64 bytes from 10.7.9.8: icmp_seq=2 ttl=64 time=7.663 ms
64 bytes from 10.7.9.8: icmp_seq=3 ttl=64 time=6.499 ms

--- 10.7.9.8 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 6.499/11.215/23.481/7.094 ms
adityasrivastava@Adityas-MacBook-Pro-103 ~ % ping -c 4 10.7.11.140
PING 10.7.11.140 (10.7.11.140): 56 data bytes
64 bytes from 10.7.11.140: icmp_seq=0 ttl=64 time=54.860 ms
64 bytes from 10.7.11.140: icmp_seq=1 ttl=64 time=11.049 ms
64 bytes from 10.7.11.140: icmp_seq=2 ttl=64 time=85.413 ms
64 bytes from 10.7.11.140: icmp_seq=3 ttl=64 time=9.610 ms

--- 10.7.11.140 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 9.610/40.233/85.413/31.799 ms
adityasrivastava@Adityas-MacBook-Pro-103 ~ % curl -si http://10.7.9.8:3001/api/status
HTTP/1.1 200 OK
X-Powered-By: Express
X-Backend: A
Cache-Control: max-age=60
Content-Type: application/json; charset=utf-8
Content-Length: 29
ETag: W/"1d-/mMMZex8nvIDi44lSUobi@vFg/M"
Date: Tue, 29 Sep 2026 11:20:18 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"backend":"A","status":"ok"}
adityasrivastava@Adityas-MacBook-Pro-103 ~ % curl -si http://10.7.11.140:3002/api/status
HTTP/1.1 200 OK
X-Powered-By: Express
X-Backend: B
Cache-Control: max-age=60
Content-Type: application/json; charset=utf-8
Content-Length: 29
ETag: W/"1d-ego46f6c2TMyh8pAIw5ntQ078PA"
Date: Tue, 29 Sep 2026 11:20:28 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"backend":"B","status":"ok"}
adityasrivastava@Adityas-MacBook-Pro-103 ~ % brew --prefix
/opt/homebrew
adityasrivastava@Adityas-MacBook-Pro-103 ~ % brew install nginx openssl@3
==> Downloading Homebrew API data
✓ JSON API packages.arm64.sequoia.jws.json Downloaded 15.6MB/ 15.6MB
Warning: nginx 1.31.6 is already installed and up-to-date.
To reinstall 1.31.6, run:
  brew reinstall nginx
openssl@3 is already installed but outdated (so it will be upgraded).
==> Downloading bottle manifests
```

```
HTTP/1.1 200 OK
X-Powered-By: Express
X-Backend: 8
Cache-Control: max-age=60
Content-Type: application/json; charset=utf-8
Content-Length: 29
ETag: W/"1d-eqo46f6c27Myh8pAIw5ntQG78PA"
Date: Tue, 29 Sep 2026 11:20:28 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"backend":"B","status":"ok"}
adityasrivastava@Adityas-MacBook-Pro-103 ~ % brew --prefix
/opt/homebrew
adityasrivastava@Adityas-MacBook-Pro-103 ~ % brew install nginx openssl@3
==> Downloading Homebrew API data
✓ JSON API packages.arm64_sequoia.json Downloaded 15.6MB/ 15.6MB
Warning: nginx 1.31.6 is already installed and up-to-date.
To reinstall 1.31.6, run:
  brew reinstall nginx
openssl@3 is already installed but outdated (so it will be upgraded).
==> Downloading bottle manifests
✓ Bottle Manifest openssl@3 (3.6.4_1) Downloaded 28.2KB/ 28.2KB
==> Would upgrade 1 formula:
openssl@3 3.6.4 -> 3.6.4_1
==> Fetching downloads for: openssl@3
✓ Bottle openssl@3 (3.6.4_1) Downloaded 11.0MB/ 11.0MB
==> Pouring openssl@3--3.6.4_1.arm64_sequoia.bottle.tar.gz
Warning: openssl@3 was installed but not linked because openssl@4 is already linked.
To link this version, run:
  brew link openssl@3
🍺 /opt/homebrew/Cellar/openssl@3/3.6.4_1: 7,639 files, 37.8MB
==> Cleanup
Removing: /opt/homebrew/Cellar/openssl@3/3.6.4... (7,639 files, 37.8MB)
Removing: /Users/adityasrivastava/Library/Caches/Homebrew/openssl@3--3.6.4... (11.0MB)
Removing: /Users/adityasrivastava/Library/Caches/Homebrew/openssl@3_bottle_manifest--3.6.4... (24KB)
Disable this behaviour by setting 'HOMEBREW_NO_INSTALL_CLEANUP=1'.
Hide these hints with 'HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').
==> Caveats
==> openssl@3
To add additional certificates, place .pem files in
/opt/homebrew/etc/openssl@3/certs

and run
/opt/homebrew/opt/openssl@3/bin/c_rehash

OpenSSL 3.6 is only supported until 2026-11-01 so the 'openssl@3'
formula will be downgraded to OpenSSL 3.5 (LTS) in a future update.

openssl@3 is keg-only, which means it was not symlinked into /opt/homebrew,
because this is an alternate version of another formula.

If you need to have openssl@3 first in your PATH, run:
  echo 'export PATH=/opt/homebrew/opt/openssl@3/bin:$PATH' >> ~/.zshrc

For compilers to find openssl@3 you may need to set:
  export LDFLAGS=-L/opt/homebrew/opt/openssl@3/lib
  export CPPFLAGS=-I/opt/homebrew/opt/openssl@3/include

The following openssl@3 executables are shadowed by other linked Homebrew commands:
  openssl (shadowed by /opt/homebrew/bin/openssl from openssl@4)
Running these by name will not invoke the version provided by this formula.
Run 'brew link openssl@3' to switch the active version to this keg.
Disable this behaviour by setting 'HOMEBREW_NO_PATH_SHADOW_CHECK=1'.
Hide these hints with 'HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').
adityasrivastava@Adityas-MacBook-Pro-103 ~ % nginx -v
nginx version: nginx/1.31.6
adityasrivastava@Adityas-MacBook-Pro-103 ~ % brew services stop nginx
```

```
For compilers to find openssl@3 you may need to set:
export LDFLAGS=-L/opt/homebrew/opt/openssl@3/lib
export CPPFLAGS=-I/opt/homebrew/opt/openssl@3/include
```

The following openssl@3 executables are shadowed by other linked Homebrew commands:

```
openssl (shadowed by /opt/homebrew/bin/openssl from openssl@4)
```

Running these by name will not invoke the version provided by this formula.

Run 'brew link openssl@3' to switch the active version to this keg.

Disable this behaviour by setting 'HOMEBREW_NO_PATH_SHADOW_CHECK=1'.

Hide these hints with 'HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % nginx -v
```

```
nginx version: nginx/1.31.6
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % brew services stop nginx
```

```
Warning: Service 'nginx' is not started.
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % cp /opt/homebrew/etc/nginx/nginx.conf /opt/homebrew/etc/nginx/nginx.conf.bak
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % mkdir -p /opt/homebrew/etc/nginx/servers /opt/homebrew/etc/nginx/certs ~/cn-edge/evidence
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % grep -n "include servers" /opt/homebrew/etc/nginx/nginx.conf
```

```
114:     include servers/*;
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo /usr/libexec/ApplicationFirewall/socketfilterfw --getglobalstate
```

```
Password:
```

```
Firewall is disabled. (State = 0)
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo /usr/libexec/ApplicationFirewall/socketfilterfw --add "${realpath /opt/homebrew/bin/nginx}"
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo /usr/libexec/ApplicationFirewall/socketfilterfw --getglobalstate
```

```
Firewall is disabled. (State = 0)
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp "${realpath /opt/homebrew/bin/nginx}"
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo /usr/libexec/ApplicationFirewall/socketfilterfw --getappblocked "${realpath /opt/homebrew/bin/nginx}"
```

```
Incoming connection to /opt/homebrew/Cellar/nginx/1.31.6/bin/nginx is permitted.
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % curl -si http://10.7.9.8:3001/api/status
```

```
HTTP/1.1 200 OK
```

```
X-Powered-By: Express
```

```
X-Backend: A
```

```
Cache-Control: max-age=60
```

```
Content-Type: application/json; charset=utf-8
```

```
Content-Length: 29
```

```
ETag: W/"1d-/mMM2ex8nvIDi44LSUobi6vFg/M"
```

```
Date: Tue, 29 Sep 2026 11:24:57 GMT
```

```
Connection: keep-alive
```

```
Keep-Alive: timeout=5
```

```
{"backend":"A","status":"ok"}
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % curl -si http://10.7.11.140:3002/api/status
```

```
HTTP/1.1 200 OK
```

```
X-Powered-By: Express
```

```
X-Backend: B
```

```
Cache-Control: max-age=60
```

```
Content-Type: application/json; charset=utf-8
```

```
Content-Length: 29
```

```
ETag: W/"1d-ego46f6cZTWyh8pAIw5ntQ078PA"
```

```
Date: Tue, 29 Sep 2026 11:25:04 GMT
```

```
Connection: keep-alive
```

```
Keep-Alive: timeout=5
```

```
{"backend":"B","status":"ok"}
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % dig @10.7.30.18 google.com +short
```

```
192.178.177.102
```

```
192.178.177.113
```

```
192.178.177.136
```

```
192.178.177.100
```

```
192.178.177.101
```

```
192.178.177.139
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo networksetup -setdnsservers Wi-Fi 10.7.30.18
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo dscacheutil -flushcache; sudo killall -HUP mDNSResponder
```

```
adityasrivastava@Adityas-MacBook-Pro-103 ~ % dig app.teamX.test
```

```
; <<> DiG 9.10.6 <<> app.teamX.test
```

```
;; global options: +cmd
```

```
;; Got answer:
```

```
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 19257
```

```
ETag: W/"1d-eqp46f6c2TMyh8pAlw5ntQG78PA"
Date: Tue, 29 Sep 2026 11:25:04 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"backend":"B","status":"ok"}
adityasrivastava@Adityas-MacBook-Pro-103 ~ % dig @10.7.30.18 google.com +short
192.178.177.102
192.178.177.113
192.178.177.138
192.178.177.100
192.178.177.101
192.178.177.139
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo networksetup -setdnsservers Wi-Fi 10.7.30.18
adityasrivastava@Adityas-MacBook-Pro-103 ~ % sudo dscacheutil -flushcache; sudo killall -HUP mDNSResponder
adityasrivastava@Adityas-MacBook-Pro-103 ~ % dig app.teamX.test

; <<>> DiG 9.10.6 <<>> app.teamX.test
;; global options: +cmd
;; Got answer:
;; -->HEADER<<- opcode: QUERY, status: NOERROR, id: 19257
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;app.teamX.test.                IN      A

;; ANSWER SECTION:
app.teamX.test.                0      IN      A      10.7.28.24

;; Query time: 88 msec
;; SERVER: 10.7.30.18#53(10.7.30.18)
;; WHEN: Tue Sep 29 16:55:58 IST 2026
;; MSG SIZE rcvd: 59

adityasrivastava@Adityas-MacBook-Pro-103 ~ % dig google.com +short
192.178.177.102
192.178.177.113
192.178.177.138
192.178.177.100
192.178.177.101
192.178.177.139
adityasrivastava@Adityas-MacBook-Pro-103 ~ % cat > /opt/homebrew/etc/nginx/servers/teamx.conf <<'EOF'
upstream backend_pool {
    server 10.7.9.8:3001 max_fails=1 fail_timeout=10s;
    server 10.7.11.140:3002 max_fails=1 fail_timeout=10s;
}

server {
    listen 80;
    server_name app.teamX.test api.teamX.test;

    location / {
        proxy_pass http://backend_pool;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_connect_timeout 2s;
        proxy_next_upstream error timeout http_502 http_503;
    }
}
EOF
adityasrivastava@Adityas-MacBook-Pro-103 ~ % cat /opt/homebrew/etc/nginx/servers/teamx.conf
upstream backend_pool {
    server 10.7.9.8:3001 max_fails=1 fail_timeout=10s;
    server 10.7.11.140:3002 max_fails=1 fail_timeout=10s;
}
```

[illegible]

[illegible]

```
nginx 20508 root 6u IPv4 0x520144f16b168ee5 0t0 TCP *:8080 (LISTEN)
nginx 20508 root 7u IPv4 0xa426cdddea687302 0t0 TCP *:80 (LISTEN)
nginx 20508 root 11u IPv4 0xb97149a8e736579b 0t0 TCP *:443 (LISTEN)
nginx 20571 nobody 6u IPv4 0x520144f16b168ee5 0t0 TCP *:8080 (LISTEN)
nginx 20571 nobody 7u IPv4 0xa426cdddea687302 0t0 TCP *:80 (LISTEN)
nginx 20571 nobody 11u IPv4 0xb97149a8e736579b 0t0 TCP *:443 (LISTEN)
adityasrivastava@Adityas-MacBook-Pro-103 certs % sudo security add-trusted-cert -d -r trustRoot -k /Library/Keychains/System.keychain /opt/homebrew/etc/nginx/certs/edge.crt
SecTrustSettingsSetTrustSettings: The authorization was canceled by the user.
adityasrivastava@Adityas-MacBook-Pro-103 certs % sudo security add-trusted-cert -d -r trustRoot -k /Library/Keychains/System.keychain /opt/homebrew/etc/nginx/certs/edge.crt
adityasrivastava@Adityas-MacBook-Pro-103 certs % security verify-cert -c /opt/homebrew/etc/nginx/certs/edge.crt -p ssl -s app.teamX.test
...certificate verification successful.
---
No extended validation result found
Certificate Transparency (CT) status: not verified
Unable to find at least 2 signed certificate timestamps (SCTs) from approved logs
adityasrivastava@Adityas-MacBook-Pro-103 certs % grep -n "upstream\|listen\|return 301\|ssl_certificate" /opt/homebrew/etc/nginx/servers/teamx.conf
1:upstream backend_pool {
7: listen 80;
9: return 301 https://$host$request_uri;
13: listen 443 ssl;
16: ssl_certificate /opt/homebrew/etc/nginx/certs/edge.crt;
17: ssl_certificate_key /opt/homebrew/etc/nginx/certs/edge.key;
27: proxy_next_upstream error timeout http_502 http_503;
adityasrivastava@Adityas-MacBook-Pro-103 certs % curl -si https://app.teamX.test/api/status
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Tue, 29 Sep 2026 11:30:52 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 29
Connection: keep-alive
X-Powered-By: Express
X-Backend: A
Cache-Control: max-age=60
ETag: W/"1d-/mMMZex8nvDI44lSUobi0vFg/M"
X-Edge: mac2

{"backend":"A","status":"ok"}
adityasrivastava@Adityas-MacBook-Pro-103 certs % curl -si https://app.teamX.test/api/status
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Tue, 29 Sep 2026 11:30:56 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 29
Connection: keep-alive
X-Powered-By: Express
X-Backend: B
Cache-Control: max-age=60
ETag: W/"1d-eqo46fec27Myh8pAlw5ntQ078PA"
X-Edge: mac2

{"backend":"B","status":"ok"}
adityasrivastava@Adityas-MacBook-Pro-103 certs % echo 'export CURL_CA_BUNDLE=/opt/homebrew/etc/nginx/certs/edge.crt' >> ~/.zshrc && so
urce ~/.zshrc
adityasrivastava@Adityas-MacBook-Pro-103 certs % curl -sI http://app.teamX.test/api/status
HTTP/1.1 301 Moved Permanently
Server: nginx/1.31.6
Date: Tue, 29 Sep 2026 11:31:09 GMT
Content-Type: text/html
Content-Length: 169
Connection: keep-alive
Location: https://app.teamx.test/api/status
adityasrivastava@Adityas-MacBook-Pro-103 certs % for i in {1..10}; do curl -sI https://app.teamX.test/api/status | grep -i x-backend; done | tee -> /cn-edge/evidence/cp7_https_roundrobin.txt
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: B
X-Backend: A
X-Backend: B
```

```
{ "backend": "A", "status": "ok" }
adityasrivastava@Adityas-MacBook-Pro-103 certs % curl -si https://app.teamX.test/api/status
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Tue, 29 Sep 2026 11:30:56 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 29
Connection: keep-alive
X-Powered-By: Express
X-Backend: B
Cache-Control: max-age=60
ETag: W/"1d-eqo46fec27Myh8pAlw5ntQ078PA"
X-Edge: mac2

adityasrivastava@Adityas-MacBook-Pro-103 certs % echo 'export CURL_CA_BUNDLE=/opt/homebrew/etc/nginx/certs/edge.crt' >> ~/.zshrc && so
urce ~/.zshrc
adityasrivastava@Adityas-MacBook-Pro-103 certs % curl -sI http://app.teamX.test/api/status
HTTP/1.1 301 Moved Permanently
Server: nginx/1.31.6
Date: Tue, 29 Sep 2026 11:31:09 GMT
Content-Type: text/html
Content-Length: 169
Connection: keep-alive
Location: https://app.teamx.test/api/status

adityasrivastava@Adityas-MacBook-Pro-103 certs % for i in {1..10}; do curl -sI https://app.teamX.test/api/status | grep -i x-backend; done | tee ~/cn-edge/evidence/cp7_https_roundrobin.txt
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
adityasrivastava@Adityas-MacBook-Pro-103 certs % curl -sI https://app.teamX.test/api/status | grep -i cache-control
Cache-Control: max-age=60
adityasrivastava@Adityas-MacBook-Pro-103 certs % open -a Safari https://app.teamX.test/api/status
adityasrivastava@Adityas-MacBook-Pro-103 certs % /opt/homebrew/etc/nginx/certs/edge.crt
zsh: permission denied: /opt/homebrew/etc/nginx/certs/edge.crt
adityasrivastava@Adityas-MacBook-Pro-103 certs % open /opt/homebrew/etc/nginx/certs
adityasrivastava@Adityas-MacBook-Pro-103 certs % grep -n -A8 -B2 "upstream backend_pool"
^C
adityasrivastava@Adityas-MacBook-Pro-103 certs % grep -n -A8 -B2 "upstream backend_pool" /opt/homebrew/etc/nginx/nginx.conf
grep -R -n -A8 -B2 "upstream backend_pool" /opt/homebrew/etc/nginx/servers/
/opt/homebrew/etc/nginx/servers/teamx.conf:1:upstream backend_pool {
/opt/homebrew/etc/nginx/servers/teamx.conf:2-     server 10.7.0.8:3001 max_fails=1 fail_timeout=10s;
/opt/homebrew/etc/nginx/servers/teamx.conf:3-     server 10.7.11.140:3002 max_fails=1 fail_timeout=10s;
/opt/homebrew/etc/nginx/servers/teamx.conf:4-}
/opt/homebrew/etc/nginx/servers/teamx.conf:5-
/opt/homebrew/etc/nginx/servers/teamx.conf:6-server {
/opt/homebrew/etc/nginx/servers/teamx.conf:7-     listen 80;
/opt/homebrew/etc/nginx/servers/teamx.conf:8-     server_name app.teamX.test api.teamX.test;
/opt/homebrew/etc/nginx/servers/teamx.conf:9-     return 301 https://$host$request_uri;
adityasrivastava@Adityas-MacBook-Pro-103 certs % for i in {1..6}; do
curl -sI https://app.teamX.test/api/status | grep -i "X-Backend:"
done
X-Backend: B
X-Backend: B
X-Backend: B
X-Backend: B
X-Backend: B
X-Backend: B
adityasrivastava@Adityas-MacBook-Pro-103 certs % [9:59 pm, 29/9/2026] Aditya Srivastava:
[10:00 pm, 29/9/2026] Aditya Srivastava: yehi aaya
[10:02 pm, 29/9/2026] Nishtha Nsy Gupta: curl -si https://app.teamX.test/api/status
zsh: bad pattern: [9:59
```

```
Server: nginx/1.31.6
Date: Tue, 29 Sep 2026 11:31:09 GMT
Content-Type: text/html
Content-Length: 169
Connection: keep-alive
Location: https://app.teamx.test/api/status

adityasrivastava@Adityas-MacBook-Pro-103 certs % for i in {1..10}; do curl -sI https://app.teamx.test/api/status | grep -i x-backend; done | tee -/cn-edge/evidence/cp7_https_roundrobin.txt
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
X-Backend: A
X-Backend: B
adityasrivastava@Adityas-MacBook-Pro-103 certs % curl -sI https://app.teamx.test/api/status | grep -i cache-control
Cache-Control: max-age=60
adityasrivastava@Adityas-MacBook-Pro-103 certs % open -a Safari https://app.teamx.test/api/status
adityasrivastava@Adityas-MacBook-Pro-103 certs % /opt/homebrew/etc/nginx/certs/edge.crt
zsh: permission denied: /opt/homebrew/etc/nginx/certs/edge.crt
adityasrivastava@Adityas-MacBook-Pro-103 certs % open /opt/homebrew/etc/nginx/certs
adityasrivastava@Adityas-MacBook-Pro-103 certs % grep -n -A8 -B2 "upstream backend_pool"
^C
adityasrivastava@Adityas-MacBook-Pro-103 certs % grep -n -A8 -B2 "upstream backend_pool" /opt/homebrew/etc/nginx/nginx.conf
grep -R -n -A8 -B2 "upstream backend_pool" /opt/homebrew/etc/nginx/servers/
/opt/homebrew/etc/nginx/servers/teamx.conf:1:upstream backend_pool {
/opt/homebrew/etc/nginx/servers/teamx.conf:2-      server 10.7.9.8:3001 max_fails=1 fail_timeout=10s;
/opt/homebrew/etc/nginx/servers/teamx.conf:3-      server 10.7.11.140:3002 max_fails=1 fail_timeout=10s;
/opt/homebrew/etc/nginx/servers/teamx.conf:4-}
/opt/homebrew/etc/nginx/servers/teamx.conf:5-
/opt/homebrew/etc/nginx/servers/teamx.conf:6-server {
/opt/homebrew/etc/nginx/servers/teamx.conf:7-      listen 80;
/opt/homebrew/etc/nginx/servers/teamx.conf:8-      server_name app.teamx.test api.teamx.test;
/opt/homebrew/etc/nginx/servers/teamx.conf:9-      return 301 https://$host$request_uri;
adityasrivastava@Adityas-MacBook-Pro-103 certs % for i in {1..6}; do
  curl -sI https://app.teamx.test/api/status | grep -i "X-Backend:"
done
X-Backend: B
X-Backend: B
X-Backend: B
X-Backend: B
X-Backend: B
X-Backend: B
adityasrivastava@Adityas-MacBook-Pro-103 certs % [9:59 pm, 29/9/2026] Aditya Srivastava:
[10:00 pm, 29/9/2026] Aditya Srivastava: yehi aaya
[10:02 pm, 29/9/2026] Nishtha Nsy Gupta: curl -sI https://app.teamx.test/api/status
zsh: bad pattern: [9:59
adityasrivastava@Adityas-MacBook-Pro-103 certs % curl -sI https://app.teamx.test/api/status
HTTP/1.1 502 Bad Gateway
Server: nginx/1.31.6
Date: Tue, 29 Sep 2026 16:32:40 GMT
Content-Type: text/html
Content-Length: 157
Connection: keep-alive
X-Edge: mac2

<html>
<head><title>502 Bad Gateway</title></head>
<body>
<center><h1>502 Bad Gateway</h1></center>
<hr><center>nginx/1.31.6</center>
</body>
</html>
adityasrivastava@Adityas-MacBook-Pro-103 certs %
```